Strength Training for Young Orienteers — Coaching Demo (~30 min)

U14 · U16 · U18 — evidence-based session plan

---

1. Why strength training for orienteers (the 2-minute "why")

Orienteers already get huge endurance stimulus from running. What running *doesn't* provide is **high-force, high-velocity neuromuscular work** — and that's exactly what improves performance and durability:

- **Better running economy** → each step on rough, hilly terrain costs less energy (Beattie et al.; Rønnestad & Mujika).
- **Injury resilience** → stronger tissues handle the eccentric load of downhill and uneven ground.
- **It's safe for youth** when supervised and technique-led — injury risk is *lower* than the sport itself (2014 International Consensus, Lloyd et al.; NSCA youth guidelines).

Core principle for the whole session: **train the movement first; load follows training age, not birth year.**

---

2. Session overview

Block	Time	Purpose
Shared dynamic warm-up	6 min	Prep, movement quality, neuromuscular activation
Group-specific main block	18 min	Strength / power scaled by training age
Shared finish + foot control	4 min	Ankle/foot, close the session

Minimal equipment: bodyweight, a few dumbbells/kettlebells (barbells for U18 if available), a box/step, a mat. Works in a hall, field, or forest clearing.

---

3. Shared dynamic warm-up (6 min)

Run continuously — also doubles as a movement-quality check for the coach.

1. Easy jog + skips, A-skip, lateral shuffle (60 s)
2. Walking lunge with reach (10 reps)
3. Inchworm to push-up position (5 reps)
4. Single-leg balance + ankle "alphabet" or hop-and-stick (8/side) — orienteering-specific
5. Pogo hops → low broad jumps, stick the landing (2×6) — introduces elastic stiffness

---

4. Group main blocks (18 min)

U14 — "Move well, land well" (low training age)

Goal: motor competence, bodyweight mastery, landing mechanics, fun. **No heavy external load.**

Circuit, 2–3 rounds, technique over reps:

- Bodyweight squat → goblet squat, light (technique focus) — 8 reps
- Split squat, bodyweight — 6/side
- Box step-up, controlled lowering — 6/side
- Calf raises, single-leg, slow eccentric lowering — 8–10 (barefoot if possible) — skill / ankle-foot control, not max load (see Calves note)
- Push-up (incline if needed) — 6–8
- Plank with shoulder taps (anti-rotation) — 20 s
- Single-leg balance reach / hop-and-stick — 5/side

Cues: "knees track over toes," "land soft and quiet," "proud chest." Keep it playful.

U16 — "Add load, own the barbell" (around/after PHV)

Goal: introduce structured external load and basic strength patterns. Manage volume if mid-growth-spurt.

Stations, 3 sets:

- Goblet or front squat (moderate, 6–8 reps, 2–3 reps in reserve)
- Romanian deadlift, DB/KB (8 reps) — posterior chain, eccentric
- Rear-foot-elevated split squat (6–8/side) — unilateral
- Single-leg calf raise, **add load** (backpack/DB), slow lowering (8–10/side) — Achilles/terrain resilience
- Pallof press or side plank (anti-rotation) — 8/side or 25 s
- Hurdle hops / depth-drop to stick (2×5) — reactive strength

Cues: brace the trunk, control the eccentric (2–3 s down), full range.

U18 — "Strong and springy" (higher training age)

Goal: maximal and explosive strength for running economy and durability.

Straight sets, 3–4 sets, longer rest (~90 s):

- Back squat or trap-bar deadlift (4–6 reps, heavy but clean, 2–3 RIR)
- Single-leg RDL or loaded split squat (6/side)
- Weighted/box jump or trap-bar jump (4–5 reps) — power
- Nordic hamstring curl or slider (5–6 reps) — eccentric injury prevention
- Heavy **loaded** single-leg calf raise (6–8/side) — push toward running-level tendon load
- Anti-rotation + anti-extension trunk combo: dead bug + Pallof (8 each)

Cues: maximal *intent* on power lifts, perfect spinal position, quality over quantity.

Progression logic to highlight live: same squat pattern scaled bodyweight → goblet → barbell across U14 → U16 → U18.

> **Calves — load it or hop it (evidence note).** A *slow bodyweight* calf raise loads the Achilles only ~1.7×BW (bilateral) to ~3.8×BW (single-leg), while running hits ~6–8×BW per stride (up to ~12.5×BW at speed), thousands of times. For athletes who already run, unloaded calf raises are a **sub-threshold strength stimulus** — one study even found moderate running's tendon strain "insufficient for adaptation." Real calf/Achilles development comes from **load** (progress to weighted single-leg raises) and **plyometrics** (the pogo/hops in the warm-up). At U14 the bodyweight version is for skill, ankle-foot control and tendon conditioning — *not* strength overload. *(Revak et al. 2017; Komi 1990/1992; Bohm et al. 2015.)*

---

5. Shared finish + foot control (4 min)

Run this as a calm guided finish, not as another training block. The goal is to close the session cleanly and reinforce ankle/foot control. This is not a stretching block, mobility work, or DOMS prevention.

1. **Settle the movement (1 min):** walk slowly around the space and shake out legs and arms.

2. **Ankle/foot control (2 min):** barefoot or thin shoes if possible.
    - **Toe yoga:** keep the ball of the foot and heel on the floor. Lift the big toe while the other toes stay down (5 reps), then keep the big toe down and lift the other toes (5 reps). Switch feet.
    - **Short foot:** keep the toes relaxed and gently draw the ball of the foot toward the heel so the arch lifts. Do not curl the toes. Hold 5 s, relax, do 5 reps/foot.
    - **Single-leg balance:** stand on one foot for 20 s/foot. Keep the base of the big toe, base of the little toe, and heel on the floor, knee tracking over toes.
3. **Close (1 min):** quick check: any sharp pain, or just normal training fatigue?

---

6. Myth-busting (the part that changes minds)

Deliver the **core three** up front (3 min); sprinkle the extras as call-outs during the practical.

Core myths

1. "Endurance athletes should stick to light, high-rep circuits." Light circuits mostly train local muscular endurance — which running already provides. The performance gains come from **heavy (≥80% 1RM, low reps) and explosive** work that fills the gap running leaves. Don't duplicate the run; complement it. *(Beattie et al.; Rønnestad & Mujika.)*

2. "Heavy and explosive lifting makes you bulky and slow." Maximal strength (1–6 reps) and explosive lifts (cleans, jumps) adapt **mostly neurally** — recruitment, rate of force development, coordination — with **minimal hypertrophy**. You get stronger *per kilo* — better strength-to-weight, exactly what an endurance athlete wants. The "bulk" fear confuses this with bodybuilding protocols (8–12 reps, high volume, to failure).

3. "Strength training is dangerous for kids / stunts growth." The growth-plate myth is **not supported by evidence**. Supervised, technique-led training is safe and beneficial, with injury rates *below* the sport itself. *(2014 International Consensus; NSCA.)*

Quick-hit extras

4. "More reps / training to failure is better." For strength and power, **intent and quality beat quantity** — heavy sets with 2–3 reps in reserve drive neural gains; grinding to failure just adds fatigue and kills bar speed.

5. "Core training = endless crunches/planks." The trunk's job in running and map-reading-on-the-move is **anti-rotation and anti-extension stiffness** to transfer force — hence Pallof presses, dead bugs, carries.

6. "Tight muscles just need stretching." What feels like "tightness" is often **protective neural tone or low stretch tolerance**, frequently tied to a **weak or unstable** muscle — not a physically short one. Stretching raises ROM mainly by making you *tolerate* it more, so it gives **temporary relief without fixing the cause** . **Full-range strength training matches stretching for flexibility** *and* addresses the weakness — and eccentric/long-length work (Nordic hamstrings, deep split squats, full-range calf raises) can even lengthen muscle structurally. For "tight" calves/hamstrings on rough terrain, loaded full range usually beats more stretching. *(Afonso et al. 2021; Alizadeh/Nuzzo et al. 2023; Weppler & Magnusson 2010; eccentric fascicle-length studies.)*

Lead message: *endurance athletes shouldn't fear heavy and explosive lifting — it won't turn them into bodybuilders, and it's the part of training the running can't replace.*

> **Case study: world-champion practice meets the science.** Orienteering World Champion **Jarkko Huovila** built his top years on heavy max-strength squats — **3×6 at 130–140 kg**, the sixth rep near-maximal ("a seventh wouldn't go"), at a **~90° knee angle (half-squat)**, year-round: **1–2×/week in winter**, dropping to maintenance **once every 7–10 days in summer**. Alongside it he did lighter muscular-endurance work (e.g. 3×60 at 60 kg) and **light deep squats for mobility** . This is essentially the **NTNU / Helgerud "maximal strength" protocol**: Støren, Helgerud, Støa & Hoff had well-trained distance runners do **half-squats, 4×4RM (~85–90% 1RM, maximal intent), 3×/week for 8 weeks** — yielding **+33% 1RM, +26% rate of force development, +5% running economy, +21% time-to-exhaustion, and no change in body weight or VO_2max** (Støren et al. 2008, *Med Sci Sports Exerc*; PMID 18460997). And head-to-head against conventional "bodybuilding"-style training at **equal volume**,

> maximal-intent training improved work economy **31% vs 18%**, 1RM **50% vs 35%** and rate of force development **155% vs 83%** — again with **no muscle-mass gain** (Heggelund et al. 2013, *Eur J Appl Physiol*; PMID 23307029). Notably, Huovila's **3×6** scheme mirrors a further NTNU study almost exactly — Hoff et al. had ~20-year-old endurance athletes do **3×6 at 85% 1RM** with maximal intent, 3×/week × 8 weeks, improving time-to-exhaustion **+20.5%** over controls via better work economy (Hoff et al. 2002, *Scand J Med Sci Sports*; PMID 12383074). Living proof of myths #1–#2: heavy, low-rep, maximal-intent lifting makes an endurance athlete *more economical and more durable without adding mass*. **Caveat:** this maximal-load protocol is a **U18+ / mature-athlete** tool — for U14/U16 it's the *direction* to build toward, not the prescription.

---

7. How to fit it into an endurance athlete's week (programming)

The honest headline: **for an endurance athlete, adding strength rarely hurts your running — it helps it.** The "interference effect" people worry about mostly works the *other way* (lots of endurance can blunt *strength/muscle* gains). Since we want **neural strength, not mass**, that interference is largely irrelevant here. So the real question isn't *whether*, it's *how to schedule it so both stay high quality*.

The four rules that matter

1. **Frequency: 2× per week is the sweet spot.** Even 1–2 short sessions maintain strength. Two 20–40 min sessions deliver almost all the benefit for a youth athlete — more isn't better once school, growth, and running load are factored in.
2. **Keep hard days hard, easy days easy.** Put strength on the **same day as a hard run**, not on recovery days. Spreading hard sessions across more days turns easy days into moderate days and erodes recovery — the most common mistake.
3. **Protect the priority of the day.** If the key *running* session is the priority, run first (fresh) and lift after, or separate by several hours. If the *strength* block is the current emphasis (e.g. off-season), lift fresh. Don't cram two maximal-quality efforts back-to-back and expect both to be good.
4. **Respect the ~24–48 h buffer before key sessions/races.** No heavy strength the day before a race or a key interval/technical orienteering session — fresh legs and clean technique come first.

Sample week (youth orienteer, ~4–6 sessions/week)

Day	Session
Mon	Easy run + mobility
Tue	**Hard:** intervals/technical orienteering → **strength** after (or 3+ h later)
Wed	Easy run or rest
Thu	Easy/technical run
Fri	**Strength** (fresh) + easy aerobic
Sat	Easy run
Sun	Long run / orienteering event

The principle: two strength sessions, both attached to days that are already "working" days, with the weekend long effort and any race protected.

Across the season (shift the emphasis, never drop it entirely)

- **Off-season / base:** best time to *build* — 2–3×/week, progress toward heavier max strength.
- **Pre-season:** *convert* to power/explosive, trim volume → 2×/week.
- **In-season / competition:** *maintain* — 1–2×/week, low volume, keep the intensity. Strength is preserved with surprisingly little work, so don't abandon it just because racing started.
- **Taper / race week:** reduce or drop heavy work 2–7 days out depending on the event's importance.

For the parents specifically

- It's **time-efficient** (2 × ~30 min) and **doesn't add injury risk** — supervised, it reduces it.
- It **doesn't make kids bulky or slow** (see myth #2) and **doesn't stunt growth** (myth #3).
- The biggest levers around it are boring but decisive: **sleep, food, and not over-scheduling.** Strength training only "works" on top of adequate recovery.

Supporting concepts: concurrent-training literature (Coffey & Hawley; Fyfe et al.); polarised load distribution / "keep easy easy" (Seiler); in-season strength maintenance (Rønnestad et al.).

---

8. DOMS — what it is and how to keep it from disrupting training

DOMS (delayed onset muscle soreness) appears **24–72 h** after unfamiliar or eccentric-heavy work. Quick myth-clear: it's **not** caused by lactic acid (that's long gone within an hour) — it's **mechanical microdamage plus an inflammatory/repair response** . A little is normal and harmless; the goal is to avoid the big, training-disrupting kind.

The one concept that explains almost everything: the repeated-bout effect

The first time you do a new or hard eccentric exercise, you get sore. Do that same exercise again and the muscle adapts fast — **soreness drops dramatically and stays low** as long as you keep training it. So DOMS is mostly a **"too much, too new, too soon"** problem, not an inherent cost of strength training.

How to actually minimise it

1. **Progress gradually.** Don't spike load, volume, *and* novel exercises at once. The most common cause of bad DOMS is too much new eccentric work in one session.
2. **Introduce new/eccentric exercises at low volume.** Nordic hamstrings are the classic offender — start with 2–3 easy reps, not 3×8. Let the repeated-bout effect build the buffer.
3. **Stay consistent.** Regular training keeps soreness minimal; long layoffs (e.g. after a break) reset the protection, so ease back in.
4. **Time novelty wisely.** Put novel exercises **early in a training block** — never in the 24–72 h window before a key interval session or competition (ties directly to the programming rules in §7).
5. **Support recovery:** sleep, enough total food and protein, and easy aerobic movement, which temporarily eases symptoms.

What does *not* reliably prevent DOMS (be honest with parents)

- **Static stretching** before or after training **does not prevent soreness** — the evidence is clear it has negligible effect (Herbert & de Noronha, Cochrane review).
- **Massage / foam rolling / cold-water immersion** can modestly reduce *perceived* soreness, but they don't fix the underlying damage — and **routine ice baths can actually blunt long-term strength adaptations** (Roberts et al.), so don't make them a habit for developing youth athletes.
- A proper **warm-up** helps performance and is worth doing, but only modestly affects DOMS.

The takeaway line for coaches/parents

Soreness is a programming signal, not a badge of honour. If a young athlete is regularly too sore to run well, the strength plan progressed too fast — scale it back, stay consistent, and the repeated-bout effect does the rest.

Supporting concepts: repeated-bout effect (McHugh); stretching ≠ DOMS prevention (Herbert & de Noronha, Cochrane); cold-water immersion blunting adaptation (Roberts et al. 2015).

9. Key sources

- Lloyd RS, et al. *Position statement on youth resistance training: the 2014 International Consensus. Br J Sports Med*, 2014; PMID 24055781.
- Lloyd RS, Oliver JL. *The Youth Physical Development Model. Strength Cond J*, 2012; DOI 10.1519/SSC.0b013e31825760ea.
- Beattie K, et al. *The effect of strength training on performance in endurance athletes. Sports Med*, 2014; PMID 24532151.
- Støren Ø, Helgerud J, Støa EM, Hoff J. *Maximal strength training improves running economy in distance runners. Med Sci Sports Exerc*, 2008; PMID 18460997.
- Heggelund J, Fimland MS, Helgerud J, Hoff J. *Maximal strength training improves work economy, rate of force development and maximal strength more than conventional strength training. Eur J Appl Physiol*, 2013; PMID 23307029.
- Hoff J, Gran A, Helgerud J. *Maximal strength training improves aerobic endurance performance. Scand J Med Sci Sports*, 2002; PMID 12383074.
- Faigenbaum AD, et al. *Youth resistance training: updated position statement paper from the National Strength and Conditioning Association. J Strength Cond Res*, 2009; PMID 19620931.
- Afonso J, et al. *Strength training versus stretching for improving range of motion: a systematic review and meta-analysis. Healthcare*, 2021; PMID 33917036.
- Alizadeh S, et al. *Resistance training induces improvements in range of motion: a systematic review and meta-analysis. Sports Med*, 2023; PMID 36622555.
- Weppler CH, Magnusson SP. *Increasing muscle extensibility: a matter of increasing length or modifying sensation? Phys Ther*, 2010; PMID 20075147.
- Coffey VG, Hawley JA. *Concurrent exercise training: do opposites distract? J Physiol*, 2017; PMID 27506998.
- Seiler S. *What is best practice for training intensity and duration distribution in endurance athletes? Int J Sports Physiol Perform*, 2010; PMID 20861519.
- Rønnestad BR, Mujika I. *Optimizing strength training for running and cycling endurance performance: a review. Scand J Med Sci Sports*, 2014; PMID 23914932.
- Roberts LA, et al. *Post-exercise cold water immersion attenuates acute anabolic signalling and long-term adaptations in muscle to strength training. J Physiol*, 2015; PMID 26174323.
- McHugh MP. *Recent advances in the understanding of the repeated bout effect: the protective effect against muscle damage from a single bout of eccentric exercise. Scand J Med Sci Sports*, 2003; PMID 12641640.
- Herbert RD, de Noronha M, Kamper SJ. *Stretching to prevent or reduce muscle soreness after exercise. Cochrane Database Syst Rev*, 2011; PMID 21735398.
- Revak A, et al. *Achilles tendon loading during heel-raising and -lowering exercises. J Athl Train*, 2017; PMID 28145739.
- Komi PV. *Relevance of in vivo force measurements to human biomechanics. J Biomech*, 1990; PMID 2081741; Komi PV, Fukashiro S, Järvinen M. *Biomechanical loading of Achilles tendon during normal locomotion. Clin Sports Med*, 1992; PMID 1638639.
- Bohm S, Mersmann F, Arampatzis A. *Human tendon adaptation in response to mechanical loading: a systematic review and meta-analysis of exercise intervention studies on healthy adults. Sports Med Open*, 2015; PMID 27747846.